

Theory of Computation



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Theory of Computation

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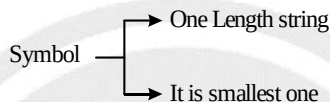


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BASICS OF THEORY OF COMPUTATION

1.1 Symbol

Symbol represents very unique in the world. Any small thing that never be broken into any other is called as symbol.



1.2 Alphabet (Σ)

It is a set of finite number of symbols.

Example:

- English alphabet = $\{a, b, \dots z\}$
- Binary alphabet = $\{0, 1\}$
- Decimal alphabet = $\{0, 1, 2, \dots 9\}$
- We can create our own alphabet = $\{\text{gate, cs, it, exam}\}$

where gate, cs, it, exam all are symbols

Alphabet (Σ) = $\{\text{gate, cs, it, exam}\}$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $\{w, x, y, z\}$

1.3 String

- It is sequence of symbols defined over given alphabet.
let $\Sigma = \{a, b\}$
- Strings possible are $\epsilon, a, b, aa, bb, \dots$
- Strings over English Alphabet: deva, gate, exam, etc.
- Strings over Binary Alphabet: 0010, 1011, 1101, 1111, etc.
- Strings over Decimal Alphabet: 3012, 2345, 5438, etc.

Note:

- Different length strings over given alphabet
 - I Zero length string $\rightarrow$ Empty string / Null string ϵ or λ is used to denote empty string length of empty string. $|\epsilon| = 0$.
 - II One length string over $\Sigma = \{a, b\}$ are a, and b (2 strings).
 - III Two length strings over $\Sigma = \{a, b\}$ are da, ab, ba, and bb (4 strings).



1.4 Operations on Strings

There are various operations on strings:

- Unary and Binary operations

1. length of a string: Length of a string is denoted as $|w|$ and is defined as the number of positions for the symbol in the string.

Example: $w = aba$
 $|w| = |aba| = 3$

2. Reversal of a string: It will reverse or changes the order of a given string w .

Example: $w = abb$
 $w^R = bba$

1.4.1 Concatenation of Two Strings

Given two strings w_1 and w_2 , we define the concatenation of w_1 and w_2 to be the string as w_1w_2 .

Example:

$w_1 = a$, and $w_2 = ba$

Then $w_1w_2 = a.ba = aba$.

1.4.2 Prefix of a String

A substring with the sequence of beginning symbols of a given string is called a “prefix”.

Example:

(i) $w = aaaa$

Prefixes = $\{\epsilon, a, aa, aaa, aaaa\}$

(ii) $w = abcd$

Prefixes = $\{\epsilon, a, ab, abc, abcd\}$

Note:

If $|w| = n$ then $|w^{Rev}| = n$.

If length of w_1 is n_1 and length of w_2 is n_2 then length of $w_1w_2 = |w_1w_2| = n_1 + n_2$.

- Let w be n length string.

(i) Number of prefixes in $w = n+1$

(ii) Number of non-empty prefixes of $w = n$

(iii) Number of different length prefixes of $w = n + 1$

(iv) Number of different length prefixes of w excluding zero length = n

1.4.3 Suffix of a String

A substring with the sequence of ending symbols of a given string is called a “suffix”.

Example:

(i) $w = aaaa$

Suffixes = $\{\epsilon, a, aa, aaa, aaaa\}$

(ii) $w = abcd$

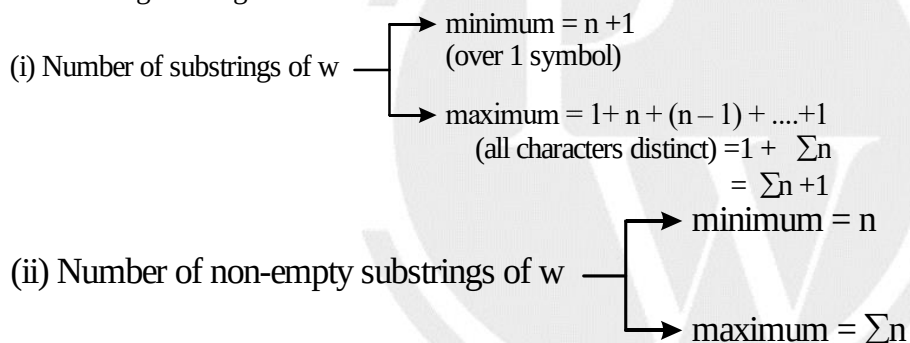
Suffixes = $\{\epsilon, d, cd, bcd, abcd\}$

Note :

- Let w be n length string.
 - (i) Number of suffixes of $w = n + 1$
 - (ii) Number of non-empty suffixes of $w = n$
 - (iii) Number of different length suffixes of $w = n + 1$
 - (iv) Number of different length suffixes of w excluding zero length = n

1.4.4 Substring of a String

- Let w be n length string.



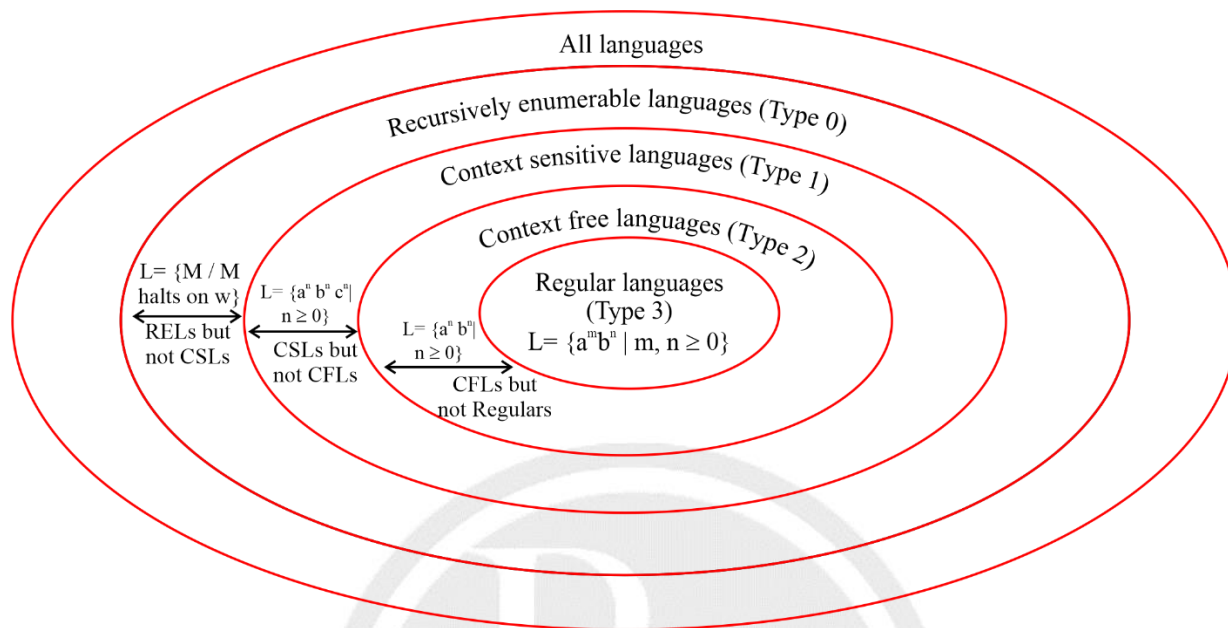
- Number of different length substrings of any given n length string = $n + 1$
- Number of different length substrings of any given n length string excluding zero length = n .

1.5 Relation between Symbol, Alphabet, String and Language

Character	Symbol
↓	↓
Character set	Alphabet
↓	↓
Token	String
↓	↓
C Language	Language

1.5.1 Chomsky Hierarchy

- Chomsky hierarchy includes all the problems in the world classified into classes.



- Type 3 is the smallest class, and Type 0 is the biggest class.

	Type 3	Type 2	Type 1	Type 0
Language:	Regular	Context free	Context sensitive	Recursively Enumerable
Automata:	Finite Automata	Push down Automata	Linear Bounded Automata	Turing Machine
Grammar:	Regular	Context free	Context Sensitive	Unrestricted

1.6 Language

- Language is set of strings defined over alphabet (Σ).
- Let $\Sigma = \{a, b\}$. Then $\Sigma = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$
 - = set of all strings
 - = universal language
 - = $\{\epsilon, a, b, aa, ab, ba, \dots\}$
- $\Sigma^2 = \Sigma^1 \cdot \Sigma^1 = \{a, b\} \cdot \{a, b\} = \{aa, ab, ba, bb\}$
- Language: It is a subset of Σ^*
- $\therefore L \subseteq \Sigma^*$
- A language is a collection of strings that must be a subset of Σ^* where Σ^* is a universal language.

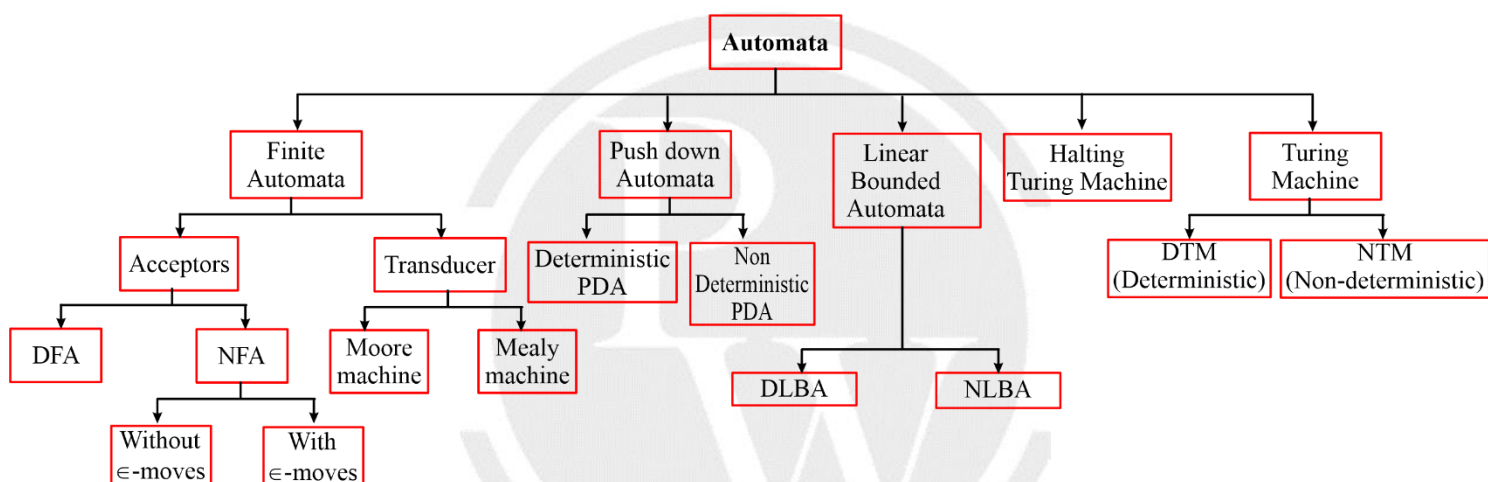
Example:

$$L = ab^* = \{a, ab, abb, \dots\}$$

1.7 Types of Languages

- Finite Language
- Infinite Language
- Regular language
- DCFL (Deterministic CFL)
- CFL
- CSL
- Recursive Language
- Recursive Enumerable Language (REL)

1.8 Types of automata



- Relations between these machines:

$$FA < DPDA < PDA < LBA < HTM < TM$$

1. **Less power:** It can Represent less number of languages.
2. **More power:** It can Represent more number of languages Compare to all these machines.

